Tristan Toshiharu Endo

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Education

University of California, San Diego — Expected June 2027
B.S. Math–Computer Science — *La Jolla, CA*

- **Relevant Coursework:** Real Analysis, Stochastic Processes, Brownian Motion, Quantitative Finance, Combinatorics, Probability Theory, Abstract Algebra

Experience

Tekly Studio, Quantitative Research & AI Engineering Lab — Jan 2026 – Present
Quantitative Researcher / AI Engineering Intern — *San Francisco, CA*

- Built and backtested ML-driven trading strategies in Python (scikit-learn, pandas) on financial time-series data, evaluating performance via Sharpe ratio and drawdown analysis
- Engineered data pipelines for preprocessing, simulation, and model evaluation, enabling rapid iteration across strategy variants

Think Twice Legal — Apr 2025 – Present
Audio Engineer / Analyst — *San Bruno, CA*

- Optimized Demucs-based source separation models via hyperparameter tuning, improving reconstruction fidelity for forensic audio analysis used in legal proceedings
- Built preprocessing and normalization pipelines for consistent model input across varying sample rates and noise conditions

Research

UC San Diego, Undergraduate Researcher — Jan 2026 – Mar 2026
Advisor: Alexander Cloninger — *La Jolla, CA*

- **Neural Network Gaussian Processes:** Derived closed-form conjugate kernels for ReLU/erf networks in Python; validated finite-width network simulations against infinite-width theoretical predictions

UC San Diego, Undergraduate Researcher — Feb 2026 – Present
Advisor: Raymond O. Chou — *La Jolla, CA*

- **Springer Fibers in Algebraic Geometry:** Built Python/SymPy toolchain enumerating Schubert cells, computing Plücker coordinates, and detecting closure relations between cells; designed an interactive 3D web visualization of the resulting cell-closure structure

Selected Projects

Springer Fiber Visualizer | *Python, JavaScript, HTML Canvas*

- Built a standalone web app rendering algebraic varieties as 3D spheres with one-directional closure-relation arrows; supports interactive first-person cell traversal with a live parameter limit display

Audio Signal Processing & ML Pipeline | *Python, Demucs* — Apr 2025 – Present

- Designed FFT-based noise reduction and stem-separation pipelines for isolating target signals in complex audio environments; built normalization workflows for consistent model input across varied audio conditions

Stochastic Simulation of Brownian Motion & OU Processes | *Python* — Sep 2025 – Dec 2025

- Simulated fractional Brownian motion across varying Hurst parameters to characterize long-range dependence; quantified empirical extrema distributions via large-scale Monte Carlo simulation

Technical Skills

Languages: Python, C, C++, Java, SQL, R, MATLAB, HTML
ML & Data: PyTorch, scikit-learn, pandas, NumPy, SymPy, FFT/signal processing
Tools: Git, Linux, Jupyter, LaTeX
Domains: Algorithmic trading, time-series modeling, stochastic processes, audio source separation, statistical modeling